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Deep learning based anomaly detection for cyber physical systems across industrial and consumer IoT environments

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Abstract

Cyber Physical Systems that support critical infrastructure are increasingly exposed to sophisticated cyber threats due to expanding connectivity and system complexity, while conventional anomaly detection approaches remain limited in scalability, adaptability to emerging attack patterns, and computational efficiency across heterogeneous environments. This study introduces a generalizable deep learning framework for anomaly detection in diverse CPS domains using a streamlined Deep Neural Network architecture that avoids complex ensemble designs while maintaining high performance. The framework integrates a structured preprocessing pipeline including normalization, categorical encoding, SMOTE based class balancing, and embedded feature selection to transform raw network traffic into discriminative inputs. Evaluation on the EdgelIoT2023 and CICIoT2023 datasets, representing industrial and consumer CPS environments, demonstrates strong cross dataset robustness rather than universal domain independence, achieving up to 99.77% accuracy and an AUC of 0.9997. Compared with contemporary boosting and hybrid models, the proposed approach delivers competitive classification capability with lower computational complexity, supporting scalable real time deployment in mission critical industrial systems and resource constrained IoT networks while establishing a foundation for adaptable and efficient CPS security solutions.

Highlights

- Lightweight AI detects cyber threats in both industrial and home IoT systems.
- High accuracy achieved without complex or resource-heavy models.
- Suitable for real-time security in low-power and edge devices.

Keywords Cyber-Physical Systems (CPS), Anomaly detection, Deep learning, Intrusion detection system (IDS), Class imbalance, SMOTE

1 Introduction

Cyber-Physical Systems (CPSs) have become integral to the functioning of critical infrastructures, spanning various sectors, including manufacturing, healthcare, energy, transportation, and agriculture. These systems combine embedded computation, real-time communication, and physical processes to enable intelligent automation and control,

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revolutionizing how industries operate and deliver services [1, 2]. The rapid evolution of CPSs has been fueled by the widespread adoption of Internet of Things (IoT) devices and the principles of Industry 4.0, resulting in highly distributed, interconnected, and adaptive environments. With advancements in sensor networks, cloud computing, and machine learning, CPSs can now make real-time decisions, perform predictive maintenance, and coordinate autonomous operations across diverse domains. While these innovations have significantly enhanced system performance, operational efficiency, and decision-making capabilities, they have also introduced new vulnerabilities. The increased interconnectivity and data exchange between devices and systems have made CPSs prime targets for cyber threats [3, 4]. Attacks targeting CPSs can result in devastating consequences, such as operational disruptions, data breaches, system failures, and even risks to human safety, making the need for robust security mechanisms critical to their continued success and resilience [5].

The challenge of securing CPSs is exacerbated by their unique characteristics. These systems often operate in heterogeneous environments, where various types of hardware, software, and communication protocols interact, making it difficult to implement a one-size-fits-all security solution [6]. CPSs are also subject to stringent real-time operational constraints, requiring security mechanisms that can operate effectively without affecting system performance. Furthermore, CPSs involve continuous interaction between cyber and physical components, which creates an inherently complex and dynamic attack surface. Traditional anomaly detection methods, such as rule-based systems and signature-based intrusion detection, have struggled to address these challenges [7, 8, 9]. These techniques are often limited by their reliance on predefined attack patterns, which are inadequate in environments where new or evolving threats emerge continuously. Moreover, these approaches often fail to adapt to the complexity of modern CPSs, especially as they scale across large, distributed networks. Conventional machine learning algorithms, such as Support Vector Machines (SVMs), Decision Trees, and Random Forests, have been applied in CPS security but face significant challenges in handling the dynamic, high-dimensional, and imbalanced data that is typical in these environments. These models often suffer from high false alarm rates, poor scalability, and reduced adaptability when confronted with unknown or zero-day attacks [10, 11]. As CPSs become more intricate and pervasive, traditional anomaly detection methods fall short in ensuring the systems' safety, highlighting the need for advanced, more efficient techniques [12, 13].

Recent advances in deep learning have shown considerable promise in addressing the challenges of CPS security, particularly in the realm of anomaly detection. Techniques such as Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs) have demonstrated the ability to capture complex spatial and temporal dependencies in data, which are essential for detecting subtle anomalies in CPS environments [14]. These models can learn intricate patterns directly from raw, high-dimensional data, reducing the reliance on extensive feature engineering and manual intervention [15, 16]. This shift toward deep learning allows for more flexible and adaptable models that can evolve with new attack strategies. Additionally, the application of advanced techniques such as attention mechanisms, temporal modeling, and unsupervised feature learning has further enhanced the ability of deep learning models to detect anomalous behaviors that may not be readily visible through traditional

statistical methods [17, 18]. These models have shown impressive performance in areas such as intrusion detection, fault diagnosis, and cybersecurity in industrial IoT systems, highlighting their potential for widespread adoption in CPSs [19].

Recent advancements in healthcare-oriented CPS environments, particularly Internet of Medical Things systems, have introduced multi attention deep CRNN architectures and blockchain integrated intrusion detection frameworks to enhance medical data protection. Studies such as XAI driven multi attention DeepCRNN models and blockchain [20, 21] assisted deep learning security frameworks demonstrate improved detection accuracy in medical IoT contexts. However, these architectures often combine recurrent, convolutional, and attention mechanisms with additional security layers, significantly increasing system complexity. While highly effective for domain specific healthcare deployments, their architectural depth and integration overhead may limit portability across heterogeneous CPS domains. The present work differs by focusing on structural simplicity and cross domain validation rather than domain specific architectural enhancement.

However, while deep learning techniques have made significant strides in CPS anomaly detection, several challenges remain. One of the primary issues is scalability [22, 23, 24, 25]. As CPSs continue to grow in size and complexity, deep learning models must be capable of handling vast amounts of data generated by sensors, devices, and systems. The high dimensionality and volume of this data can overwhelm existing models, leading to performance degradation and increased computational costs [26, 27]. Furthermore, many deep learning models used in CPS environments are still domain-specific, which limits their ability to generalize across different sectors and use cases. For example, models trained on industrial CPS data may struggle to adapt to consumer IoT systems, where attack patterns and data characteristics may differ significantly. Another challenge is the issue of class imbalance, as malicious activities often represent only a small fraction of the total data in CPS environments [28, 29]. Although techniques like Synthetic Minority Over-sampling Technique (SMOTE) and hybrid approaches have been introduced to address this issue, they are not always effective in all scenarios [26, 30]. Additionally, many existing models rely on complex ensemble architectures, which, while improving performance, introduce significant computational overhead and reduce the model's efficiency [31, 32]. These challenges underscore the need for more scalable, flexible, and efficient deep learning-based anomaly detection frameworks that can operate effectively across a wide range of CPS environments.

Furthermore, the integration of Information Technology (IT) and Operational Technology (OT) in CPSs introduces additional complexities. The convergence of these two domains expands the attack surface, making CPSs more susceptible to multi-stage attacks that exploit both digital vulnerabilities and physical processes [33, 34]. As CPSs increasingly connect with external networks, they are exposed to a broader array of cyber threats, many of which target both the digital control systems and the physical assets they manage. This fusion of IT and OT infrastructures has led to the emergence of sophisticated cyber-attacks that exploit both the physical and cyber components of a system [35, 36]. While recent deep learning models have made some progress in detecting these types of attacks, the real-time nature of CPSs and their continuous interaction between cyber and physical layers complicates detection and mitigation efforts. The detection of these multifaceted threats requires advanced models that can process large

amounts of heterogeneous data in real-time while accurately distinguishing between benign and malicious activities. Addressing these challenges requires the development of models that are capable of handling high-dimensional and real-time data while demonstrating robust performance across selected heterogeneous CPS domains.

The rapid expansion of Cyber Physical Systems across industrial control environments and consumer Internet of Things networks has intensified the demand for scalable and context aware cybersecurity solutions. As interconnectivity increases, these infrastructures become more vulnerable to sophisticated and evolving cyber-attacks. Although recent research has reported high detection accuracy using CNN LSTM hybrids, dual attention mechanisms, and transformer-based encoders, many of these architectures are validated within single domain datasets. When transferred to different CPS environments, they often require extensive hyperparameter tuning or architectural modification [37]. Moreover, recurrent and attention-based modules increase parameter volume, training instability, and inference latency, which limits practical deployment on edge and gateway devices. Much of the existing literature emphasizes architectural novelty rather than operational portability and evaluated cross-domain robustness across selected CPS datasets.

To address related challenges, several contemporary approaches have incorporated hybrid blockchain integrated intrusion detection frameworks, federated learning strategies, transfer learning based BiLSTM models, and explainable deep learning mechanisms tailored for Industry 5.0 environments. For instance, GAO XGBoost combined with elliptic curve cryptography blockchain mechanisms enhances data integrity, while federated and blockchain integrated IDS frameworks focus on privacy preservation and decentralized coordination. Transfer learning BiLSTM architectures aim to improve adaptability across IoT domains, and Alpha Net proposes trustworthy deep learning for industrial botnet detection. In contrast to these cryptographic or distributed trust-oriented frameworks, the present study concentrates on lightweight operational security. Rather than integrating additional infrastructure layers, it prioritizes architectural restraint, computational efficiency, and validated cross dataset robustness.

In this study, *cross-domain generalization* is defined as the ability of a model to maintain consistent detection performance across multiple, structurally heterogeneous Cyber-Physical System (CPS) datasets evaluated under a unified architectural and training configuration. Specifically, this work considers two distinct CPS domains: industrial edge environments (EdgeIIoT2023) and consumer IoT ecosystems (CICIoT2023). Therefore, generalization is assessed as performance stability across these selected domains, rather than as universal applicability across all possible CPS environments. This distinction is important, as CPS deployments vary widely in protocols, operational constraints, and attack surfaces, and broader generalization beyond the evaluated domains is not assumed.

Motivated by these limitations, this research investigates how a scalable and efficient anomaly detection framework can generalize across selected heterogeneous CPS domains (industrial and consumer IoT) while maintaining high detection performance. Instead of introducing complex hybrid or ensemble structures, the study demonstrates that a carefully optimized fully connected deep neural network with controlled depth and parameter count can achieve strong cross domain performance.

While traditional shallow machine learning models such as Support Vector Machines and Random Forests offer low computational complexity, they rely heavily on fixed feature representations and limited hierarchical abstraction. In contrast, even a minimalist Deep Neural Network (DNN) provides multi-level nonlinear feature transformation, enabling the model to learn progressively abstract representations of CPS traffic patterns. This capability is particularly important in cross-domain settings, where statistical distributions and attack signatures differ across environments. Unlike shallow models, which depend on manually engineered or static feature spaces, the DNN can adapt its internal feature representations through training, improving its ability to capture domain-invariant patterns across heterogeneous datasets. Therefore, the choice of a lightweight fully connected DNN represents a balance between representational capacity and computational efficiency, making it more suitable for cross-domain generalization than traditional shallow models.

The architecture relies exclusively on dense layers with explicit regularization and disciplined dimensionality reduction, avoiding attention mechanisms, recurrent gating structures, temporal convolutional blocks, and ensemble aggregation. This structural simplicity reduces parameter redundancy, improves training stability, and lowers computational overhead, thereby supporting deployment on resource constrained edge devices common in industrial and consumer IoT settings.

The contribution therefore lies not merely in employing a single model, but in validating that architectural minimalism combined with disciplined preprocessing can sustain competitive detection capability across structurally different CPS datasets without domain-specific redesign. In this study, generalization refers specifically to cross-domain generalization across the two evaluated CPS datasets EdgeIIoT2023 and CICIoT2023 under a unified architectural design and training protocol. These datasets represent distinct operational environments, including industrial edge systems and consumer IoT ecosystems. Therefore, the observed performance consistency provides empirical evidence of generalization across selected heterogeneous domains, rather than universal applicability. This study does not claim universal domain independence across all CPS environments. Broader generalization beyond these domains requires further validation using additional datasets and real-world deployments.

2 Methodology

This section outlines the methodology employed to design, develop, and evaluate a Deep Neural Network (DNN)-based anomaly detection framework tailored for Cyber-Physical Systems (CPSs). The goal is to create a transparent, reproducible, and scalable model capable of effectively detecting anomalous behaviors in complex CPS environments. The process involves six key stages: dataset acquisition, data preprocessing, class balancing, feature selection, model construction and training, and performance evaluation.

In the dataset acquisition phase, publicly available network traffic datasets relevant to CPS and IoT infrastructures are chosen. These datasets contain labeled examples of both benign and malicious traffic, offering a solid foundation for supervised learning. Special emphasis is placed on datasets that reflect the heterogeneity, protocol diversity, and real-time characteristics common in operational CPS networks, ensuring that the data represent a broad spectrum of scenarios. The data preprocessing stage transforms the raw dataset into a machine-readable format. This includes cleaning the data by removing

missing values, normalizing continuous variables, and encoding categorical features to ensure consistency. Additionally, irrelevant or noisy attributes are filtered out, and standardization techniques, such as min-max normalization, are applied to ensure all features contribute equally to the learning process.

Addressing the class imbalance issue, where benign instances vastly outnumber malicious ones, is a critical step in developing an effective anomaly detection framework. This imbalance can lead to a model's inability to detect rare or infrequent attacks, thus reducing its overall performance. To tackle this challenge, the Synthetic Minority Oversampling Technique (SMOTE) is employed to generate synthetic examples of the minority class instances. By augmenting the minority class, SMOTE enhances the model's sensitivity to rare attacks and reduces the risk of bias during training, enabling the model to effectively learn and identify even infrequent anomalies.

It is important to recognize that CPS network traffic exhibits temporal correlations, and applying synthetic oversampling may distort the sequential structure of the data. To mitigate this issue, SMOTE is applied after temporal aggregation features are extracted. Temporal aggregation was performed to transform sequential network traffic observations into statistically stable feature representations before the application of oversampling. Specifically, raw packet-level and flow-level observations were aggregated using fixed time windows, within which statistical descriptors such as mean packet size, packet inter-arrival time variance, byte transfer rate, protocol frequency distribution, and connection duration statistics were computed. These aggregated descriptors capture the temporal behavior of network activity while reducing sensitivity to short-term fluctuations. Applying SMOTE after this aggregation stage ensures that synthetic samples are generated within the derived statistical feature space rather than the raw sequential traffic space. Consequently, the synthetic samples preserve the statistical relationships among traffic attributes while minimizing distortion of underlying temporal patterns. This strategy enables the use of a feed-forward DNN architecture by encoding temporal dynamics into statistical feature representations during the aggregation stage. It is important to note that the DNN itself does not explicitly model temporal dependencies; rather, it operates on temporally aggregated feature vectors, treating each instance as independent. Thus, temporal behavior is captured during feature engineering, while the DNN performs classification on the resulting static representations.

This approach reduces the risk of disrupting raw time dependencies while still addressing the class imbalance. During preliminary experiments, alternative imbalance strategies, such as class weighted loss and focal loss, were also considered. However, SMOTE was found to provide more stable convergence and significantly improve recall for the minority class, making it the preferred approach for this study. While SMOTE proves effective in improving model performance in imbalanced datasets, the limitations of synthetic sampling in dynamic CPS contexts remain a consideration. The potential for synthetic data to introduce artifacts or fail to capture the full complexity of real-world attack patterns necessitates further research and caution in its application.

The feature selection phase aims to reduce the dataset's dimensionality by retaining only the most relevant features. eature selection was implemented as a two-stage preprocessing pipeline combining filter-based and wrapper-based methods. First, mutual information (MI) analysis was applied to rank all features according to their statistical dependency with the target labels. Based on this ranking, a subset of top-ranked features

was selected as input to the second stage. Subsequently, recursive feature elimination (RFE) was applied using a logistic regression classifier as the base estimator to iteratively remove the least informative features. During this process, feature importance was evaluated using cross-validated classification performance, and features were recursively pruned until an optimal subset was obtained. The final selection of 32 features was determined based on the point at which further feature removal resulted in no significant improvement or slight degradation in validation performance. This hybrid MI–RFE strategy balances computational efficiency with predictive relevance, ensuring that the selected features are both statistically informative and model-relevant.

During the model construction and training stage, a deep neural network is designed to learn patterns from the preprocessed and balanced data. The network consists of multiple fully connected layers with ReLU activation functions, dropout layers for regularization, and a sigmoid output layer for binary classification. Training is performed using the Adam optimizer and binary cross-entropy loss function. Hyperparameter tuning is carried out through grid search and cross-validation to ensure optimal model performance.

The hyperparameters explored during grid search included learning rate (0.0001, 0.0005, 0.001), batch size (32, 64, 128), number of hidden layers (2–4), number of neurons per layer (32–128), dropout rate (0.2–0.5), and L2 regularization coefficient (0.001–0.1). Model selection was based on validation AUC and F1-score, with early stopping applied to prevent overfitting. The optimal configuration was chosen as the model achieving the best trade-off between classification performance and training stability across cross-validation folds.

Finally, in the performance evaluation phase, the model's effectiveness is assessed using a variety of standard metrics, including accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). These metrics provide a thorough evaluation of the model's classification capabilities, particularly its efficiency in identifying malicious traffic without generating excessive false positives.

Overall, this methodology establishes a comprehensive and systematic approach to developing a DNN-based anomaly detection framework for CPSs. Through careful dataset selection, preprocessing, and feature engineering, followed by rigorous model construction and evaluation, the framework is designed to be both scalable and efficient, offering high detection performance in diverse and complex CPS environments.

Feature selection was implemented as a separate preprocessing stage using mutual information ranking followed by recursive feature elimination. The top 32 features were selected based on cross validated importance scores. Although not embedded within the backpropagation process itself, the selected features directly constrain the DNN input space, effectively embedding dimensional control into the training pipeline. This reduction from the original feature sets significantly decreased parameter count and reduced training time while maintaining classification performance.

2.1 System architecture

The proposed anomaly detection framework is organized into a modular, three-stage architecture: data collection, preprocessing and learning, and anomaly detection output. This design reflects the operational pipeline of real-world Cyber-Physical Systems (CPS) monitoring and threat detection systems. In the data collection stage, raw network

traffic data is gathered from CPS environments. This data can include both flow-level and packet-level information such as IP addresses, ports, protocol types, timestamps, byte counts, and traffic direction. Such data provides a comprehensive foundation for training and inference, capturing the operational behavior of the CPS and acting as the primary source of input for anomaly detection. The quality and variety of data collected in this stage ensure that the system can learn to differentiate between normal and anomalous behaviors in different CPS contexts.

The collected data then moves into the preprocessing and learning stage, where it undergoes several critical transformations to prepare it for use in the model. First, the data is cleaned by removing noise and handling missing values, ensuring that only valid and relevant information is retained. Continuous features are then scaled, and categorical variables are transformed using appropriate encoding techniques, making the data suitable for processing by machine learning models. A key component of this stage is the application of the Synthetic Minority Over-sampling Technique (SMOTE) to address class imbalance. In most CPS scenarios, benign traffic vastly outnumbers malicious traffic, which can lead to biased model training. SMOTE generates synthetic instances of the minority class (malicious traffic), ensuring that the model learns to recognize rare attack types. Once the data is cleaned, normalized, and balanced, it is fed into a multilayer Deep Neural Network (DNN). The DNN is designed to extract high-level feature representations and learn the mapping between input features and class labels (benign or malicious). The DNN architecture is optimized for both efficiency and accuracy, using non-linear transformations to capture complex patterns inherent in network traffic data.

To ensure full reproducibility and transparent comparison with existing anomaly detection models, the architectural configuration of the proposed deep neural networks is summarized in Table 1. Although both models follow the same design philosophy of lightweight fully connected networks with controlled depth, slight architectural adjustments were introduced to accommodate the statistical characteristics of each dataset. The architectures were determined through systematic hyperparameter optimization using grid search, balancing classification performance, training stability, and computational efficiency. Key design parameters include the number of hidden layers, neuron counts, activation functions, regularization strategies, dropout rates, and optimization settings. The final configurations maintain low parameter counts while preserving strong discriminative capability for CPS network traffic.

The final optimized architectures are as follows.

Table 1 Detailed architectural configuration of the proposed DNN models for EdgeIIoT2023 and CICIoT2023 datasets

Component	EdgeIIoT2023 Model	CICIoT2023 Model
Input Features	32	32
Hidden Layer 1	64 neurons (ReLU)	64 neurons (ReLU)
Hidden Layer 2	32 neurons (ReLU)	32 neurons (ReLU)
Hidden Layer 3	64 neurons (ReLU)	—
Dropout	0.4	0.2
Regularization	L2 ($\lambda = 0.1$)	L2 ($\lambda = 0.01$)
Output Layer	1 neuron (Sigmoid)	1 neuron (Sigmoid)
Optimizer	Adam	Adam
Loss Function	Binary Cross-Entropy	Binary Cross-Entropy
Batch Size	64	64
Training Epochs	30 (early stopping)	20 (early stopping)

For EdgeIIoT2023:
Input layer: 32 features
Dense layer 1: 64 neurons, ReLU activation
Dense layer 2: 32 neurons, ReLU activation
Dense layer 3: 64 neurons, ReLU activation
Dropout: 0.4
L2 regularization coefficient: 0.1
Output layer: 1 neuron, Sigmoid activation
For CICIoT2023:
Input layer: 32 features
Dense layer 1: 64 neurons, ReLU activation
Dense layer 2: 32 neurons, ReLU activation
Dropout: 0.2
L2 regularization coefficient: 0.01
Output layer: 1 neuron, Sigmoid activation
These configurations were selected through grid search based on validation AUC and stability criteria.

Hyperparameters explored during grid search included learning rate values of 0.0001, 0.0005, and 0.001; batch sizes of 32, 64, and 128; hidden layer widths between 32 and 128 neurons; network depth between two and four layers; dropout rates between 0.2 and 0.5; and L2 regularization coefficients between 0.001 and 0.1. The selected configurations achieved optimal balance between validation performance and computational efficiency.

In the final anomaly detection output stage, the trained DNN processes incoming data to determine whether each instance represents normal or anomalous behavior. The output layer of the network provides a binary classification, indicating whether the observed behavior is benign or malicious. This enables the system to detect and classify suspicious activities in real time. The modular structure of this architecture ensures that it can be seamlessly integrated with existing CPS intrusion detection and monitoring platforms, allowing for proactive and timely threat response with minimal latency. The three-stage architecture not only provides a scalable and extensible framework for anomaly detection in CPS environments but also offers flexibility for future upgrades. For instance, additional security modules or more complex network architectures can be incorporated as the system evolves. As depicted in Fig. 1, the architecture reflects the logical flow of data—from raw input acquisition to real-time threat classification—making it adaptable to various CPS applications.

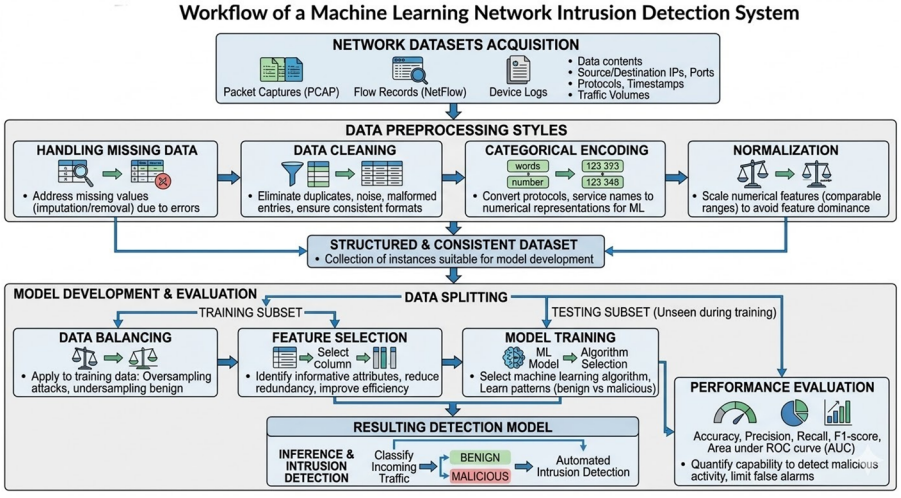


Fig. 1 Proposed System Architecture

2.2 Datasets collection

To ensure comprehensive coverage of heterogeneous CPS scenarios, this study utilizes two publicly available, high-quality datasets: EdgeIIoT2023 and CICIoT2023. These datasets were chosen for their relevance to real-world CPS applications, their diversity in attack types, and their extensive, well-labeled network traffic data, including both benign and malicious traffic. EdgeIIoT2023 is derived from a smart factory testbed designed to simulate edge-computing CPS environments. This dataset includes a broad set of attack scenarios common in industrial contexts, such as Distributed Denial of Service (DDoS), spoofing, Man-in-the-Middle (MITM) attacks, data injection, and malware propagation. It mirrors the low-latency, high-throughput characteristics that are typical of edge-enabled CPS environments, making it an ideal source for training models aimed at industrial applications. The dataset captures both routine network traffic and attack-induced behaviors, providing a realistic foundation for developing and evaluating security models under operational conditions that closely resemble real-world industrial systems.

In contrast, CICIoT2023, developed by the Canadian Institute for Cybersecurity, focuses on smart home and consumer IoT environments. This dataset contains labeled traffic generated by real devices such as IP cameras, smart thermostats, and plugs. The dataset covers a wide range of attack types, including brute-force attempts, cross-site scripting (XSS), infiltration, and SQL injection, which are particularly relevant to the vulnerabilities found in home IoT systems. CICIoT2023 also incorporates time-series and statistical features, which are critical for deep learning applications in temporal anomaly detection. This aspect of the dataset makes it well-suited for capturing the dynamic nature of IoT traffic and detecting anomalies based on both historical and real-time data streams.

Together, the datasets EdgeIIoT2023 and CICIoT2023 offer a broad spectrum of CPS scenarios, representing both industrial edge environments and consumer IoT systems. By utilizing these datasets, the proposed framework is exposed to a diverse set of traffic patterns, attack modalities, and operational characteristics, supporting evaluation of generalization across two representative and structurally distinct CPS domains. This dual-dataset approach also enhances the robustness of the model, as it is trained on data that reflects the varied security needs of different types of CPS. The inclusion of both industrial and consumer IoT environments ensures that the framework is adaptable to various attack vectors and network behaviors commonly found in modern CPS deployments. As a result, the datasets provide an ideal basis for evaluating the effectiveness of the proposed anomaly detection framework in real-world CPS scenarios.

The selection of EdgeIIoT2023 and CICIoT2023 was also motivated by several limitations observed in earlier benchmark intrusion detection datasets such as NSL-KDD, UNSW-NB15, and Bot-IoT. Many legacy datasets contain outdated attack signatures, simplified traffic patterns, or limited protocol diversity, which may not accurately represent the dynamic behavior of modern CPS infrastructures. In contrast, EdgeIIoT2023 captures industrial edge-computing traffic generated from smart manufacturing testbeds, including industrial control communication patterns and operational latency constraints. Meanwhile, CICIoT2023 reflects consumer-oriented IoT ecosystems consisting of heterogeneous smart devices operating under realistic home network conditions. The combination of these datasets therefore enables evaluation across two distinct CPS

operational domain industrial infrastructure and consumer IoT networks providing a meaningful testbed for assessing cross-dataset robustness and architectural adaptability.

2.2.1 Data preprocessing

The EdgeIIoT2023 and CICIoT2023 datasets were generated in controlled laboratory environments, where normal and malicious traffic patterns were synthetically created to reflect plausible real-world behavior. These datasets include labeled instances critical for supervised learning where benign and attack data are explicitly distinguished, allowing the model to learn patterns associated with anomalous activity.

To ensure data quality and compatibility with the model, a rigorous preprocessing pipeline was implemented. The first step involved data cleaning, where duplicate records were removed specifically, 815 entries from the EdgeIIoT2023 dataset and 6 entries from the CICIoT2023 dataset. This resulted in final dataset sizes of 2,218,386 and 2,867,727 instances, respectively. Both datasets were free from missing values, so no imputation was required. Next, categorical encoding was applied to transform non-numeric attributes into numerical representations suitable for neural networks. In EdgeIIoT2023, 20 categorical features were encoded, expanding the feature space to a total of 89 features. Similarly, CICIoT2023 resulted in 62 final features after encoding. Subsequently, all numerical attributes underwent Min-Max normalization, which scaled their values to the [0, 1] range. This step standardized feature distributions and helped expedite model convergence. Notably, Min-Max normalization was applied exclusively to continuous features, while z-score standardization was avoided, ensuring that feature magnitudes remained consistent across datasets while preserving their inherent distributional relationships.

To prevent data leakage and ensure unbiased performance estimation, dataset splitting was performed before any preprocessing operations. Feature selection and SMOTE oversampling were applied exclusively to the training subset, leaving the validation and test sets untouched. This approach ensured that the validation and test sets remained independent, maintaining the integrity of performance evaluation.

2.2.2 Training and testing dataset splitting

To facilitate model evaluation and mitigate data leakage, each dataset was partitioned into training and testing subsets using an 80:20 split. As shown in Fig. 2, stratified sampling was employed to preserve the original class distribution across both subsets. This ensures that both benign and malicious traffic patterns are proportionally represented during training and evaluation. Class labels were standardized by mapping benign samples to 0 and malicious samples to 1. A fixed random seed was used to ensure reproducibility of the data split across multiple experiments.

2.2.3 Dataset balancing

An inherent limitation of both datasets is the class imbalance, with benign traffic overwhelmingly dominating the An inherent limitation of both the EdgeIIoT2023 and CICIoT2023 datasets is the significant class imbalance, where benign traffic overwhelmingly dominates the minority malicious samples. To address this issue, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training sets. SMOTE generates synthetic instances by interpolating between existing minority class samples,

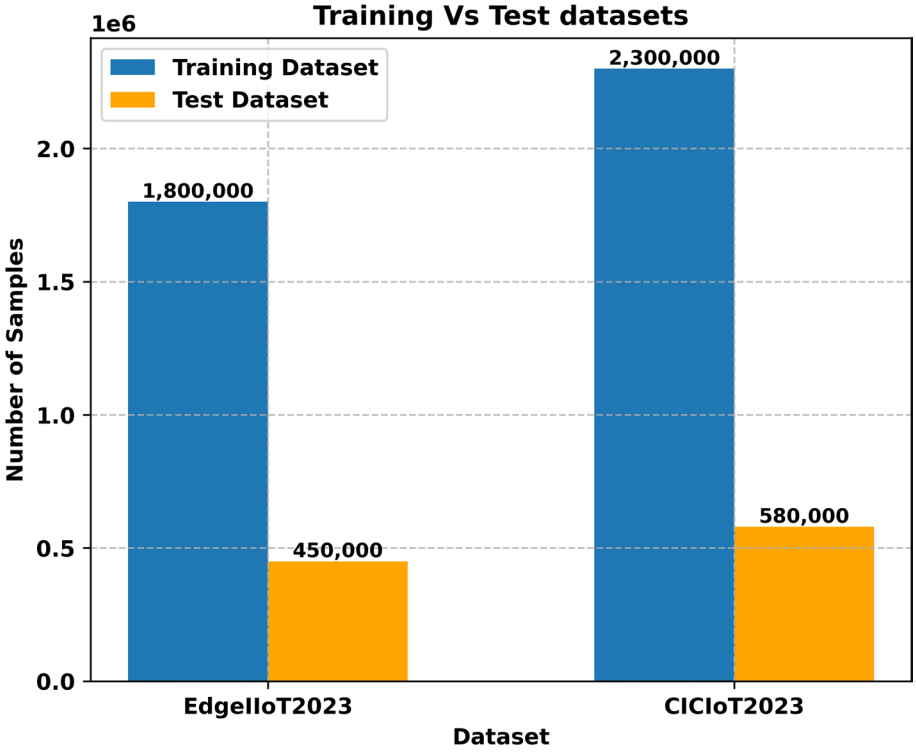


Fig. 2 Training and testing datasets

thereby improving class balance without simply replicating original data. This method increases the exposure of the minority class during training by generating feature-consistent malicious samples, allowing the model to better capture rare attack behaviors. The goal is not just to equalize class counts, but to enhance the model's ability to learn decision boundaries for minority class anomalies. By increasing the representation of malicious traffic in the training space, SMOTE reduces bias towards benign predictions, improving recall and reducing false negative rates during evaluation. The effectiveness of this strategy is later assessed through recall and false negative rate analysis in the Results section to examine its impact on detecting minority class attacks.

Following the balancing procedure, the datasets underwent feature selection to reduce dimensionality and computational overhead while preserving discriminative power. Feature selection was implemented through an embedded deep learning–based bottleneck extraction mechanism integrated within the DNN architecture. In this approach, an intermediate bottleneck layer compresses the original feature representation into a lower-dimensional latent space, forcing the network to retain only the most discriminative information necessary for classification. During supervised training, the network learns weight parameters that map the high-dimensional input space to a compact representation guided by the class labels. The importance of each feature is implicitly determined through its contribution to the learned latent representation. Following convergence, the output of the bottleneck layer is extracted and used as the reduced feature set. This process resulted in the selection of 32 informative features for both datasets. Compared with traditional filter-based or wrapper-based selection methods, this embedded approach simultaneously optimizes feature relevance and classification performance while minimizing computational overhead.

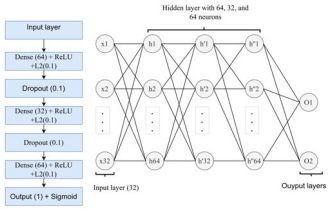


Fig. 3 DNN model Architecture for the EdgeIIoT2023 Dataset

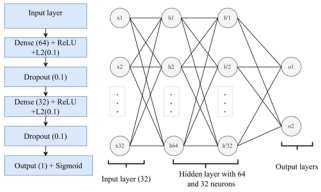


Fig. 4 DNN model Architecture for the CICIoT2023 Dataset

2.2.4 Tailored DNN architectures and training strategies for EdgeIIoT2023 and CICIoT2023

For the final classification task, two tailored Deep Neural Network (DNN) architectures were developed, each optimized for its respective dataset. These architectures were trained on data that had been balanced using the Synthetic Minority Over-sampling Technique (SMOTE) and reduced through feature selection to ensure the models were both efficient and effective. Both models leveraged the Adam optimizer and binary cross-entropy loss function, with early stopping incorporated to prevent overfitting during training. A validation split was maintained throughout training to monitor performance on unseen data and ensure that the models generalized well to new, unseen scenarios.

For the EdgeIIoT2023 dataset, the DNN architecture, as illustrated in Fig. 3, consists of an input layer with 32 features, followed by three fully connected hidden layers containing 64, 32, and 64 neurons, respectively. Each of these layers employed the ReLU (Rectified Linear Unit) activation function, which is commonly used for its ability to introduce non-linearity into the model and help it learn complex patterns. Additionally, the network was regularized using L2 regularization with a penalty coefficient of $\lambda = 0.1$ to prevent overfitting by penalizing large weights. A dropout layer was included after the dense layers with a rate of 0.4 to further reduce overfitting by randomly dropping neurons during training, ensuring the model doesn't rely too heavily on specific features or paths. The final output layer used a sigmoid activation function to perform binary classification, distinguishing between benign and malicious traffic. The model was trained for a maximum of 30 epochs, with early stopping triggered by a plateau in the validation loss to avoid unnecessary overfitting.

In contrast, the DNN architecture for the CICIoT2023 dataset, shown in Fig. 4, featured a more lightweight design with two hidden layers. The input layer accepted 32 features, similar to the EdgeIIoT2023 model. The first hidden layer consisted of 64 neurons, and the second hidden layer contained 32 neurons. Both hidden layers utilized the ReLU activation function and were regularized with L2 regularization ($\lambda = 0.01$) to avoid overfitting. Each dense layer was followed by a dropout layer with a rate of 0.2 to promote robustness and prevent overfitting by randomly omitting certain units during training. The output layer was a single neuron using a sigmoid activation function to classify the

instances as either benign or malicious. Training was carried out for 20 epochs, using a batch size of 64, and incorporated a 20% validation split with early stopping set at a patience parameter of 5 to halt training when validation performance ceased improving.

2.2.5 Data preprocessing: SMOTE and feature selection

To address the inherent class imbalance present in the datasets, where benign traffic greatly outnumbers malicious traffic, SMOTE was employed. SMOTE is particularly effective for anomaly detection, as it generates synthetic instances of the minority class (malicious traffic), thereby balancing the dataset. This method helps the model become more sensitive to rare attack patterns, as it effectively augments underrepresented attack classes. By synthesizing realistic data points for underrepresented attack types, SMOTE enhances the model's ability to detect anomalous behaviors, which is crucial for identifying emerging or previously unseen threats in CPS environments.

Alongside the application of SMOTE for class balancing, feature selection techniques were employed to reduce the dimensionality of the dataset and eliminate irrelevant or redundant features. This not only improves the model's efficiency by reducing the number of parameters but also enhances interpretability by focusing on the most relevant attributes. Feature selection ensures that the model learns from the most significant and informative features, which in turn improves overall performance and enables better generalization across diverse CPS environments. By reducing overfitting and preventing the model from learning noise or irrelevant patterns, feature selection plays a crucial role in enhancing the robustness and accuracy of the final model.

Interpretability in this study is supported through feature importance ranking derived from mutual information analysis. By constraining the model to 32 informative features, the decision space is made more traceable compared to high-dimensional hybrid networks. This approach facilitates a clearer understanding of the model's decision-making process. Future work will integrate SHAP-based attribution analysis to provide instance-level interpretability and quantify the contribution of each feature during prediction, offering a deeper insight into the model's behavior.

Algorithm 1 Deep Neural Network Feature Selection via Bottleneck Extraction.

Input: High-variance training data X_train $\in \mathbb{R}^{N\times D}$, test data X_test $\in \mathbb{R}^{M\times D}$, labels y_train $\in \{0,1\}^N$
Output: Reduced features X_train_reduced $\in \mathbb{R}^{N\times K}$, X_test_reduced $\in \mathbb{R}^{M\times K\ (K=32)}$

```
Set input_dim ← D                    // Number of input features
Define DNN model:
    inputs ← Input(shape = (D,))
    h1 ← Dense(64, ReLU)(inputs)
    bottleneck ← Dense(K, ReLU, name = "automatic_features_layer")(h1)
    h2 ← Dense(64, ReLU)(bottleneck)
    output ← Dense(1, sigmoid)(h2)
    model ← Model(inputs, output)
Build model: dummy ← random((1, D)); model(dummy)     // Initialize weights
Train model on (X_train, y_train) using binary cross-entropy loss     // Supervised pretraining
Create extractor:
    features_model ← Model(model.input, model.get_layer("automatic_features_layer").output)
X_train_reduced ← features_model.predict(X_train)
X_test_reduced ← features_model.predict(X_test)
return X_train_reduced, X_test_reduced
```

The algorithm describes a supervised deep neural network used not only for classification, but also for automatic feature reduction through a bottleneck layer.

The input to the algorithm is high-variance training data $X_{train} \in \mathbb{R}^{N \times D}$, test data $X_{test} \in \mathbb{R}^{M \times D}$, and binary labels $y_{train} \in \{0,1\}^{N}$. In the code, this corresponds to X_train_high_variance, X_test_high_variance, and y_train. The dimensionality D is obtained dynamically using DNN architecture.

The layer named "automatic_features_layer" corresponds exactly to the bottleneck in the algorithm. This layer compresses the representation from 64 dimensions down to 32, forcing the network to learn a compact set of discriminative features. Because the network is trained in a supervised manner, this compression is guided by the class labels, which typically leads to more task-relevant features than unsupervised dimensionality reduction methods. Step 3 of the algorithm initializes the model weights by passing a dummy input through the network.

This ensures that the model is built and all weights are initialized before extracting intermediate layers.

Step 4 in the algorithm involves training the model using binary cross-entropy loss on (X_train, y_train). This supervised training phase is essential, because the bottleneck features become meaningful only after the network has learned to separate the classes.

Step 5 creates a new model that maps the original inputs directly to the output of the bottleneck layer. This intermediate model discards the final classification layers and retains only the mapping up to the bottleneck. Conceptually, this transforms the DNN from a classifier into a feature extractor.

Finally, Steps 6 and 7 apply this feature extractor to both training and test data:

The result is:

$$X_{train_reduced} \in \mathbb{R}^{N \times 32}$$

$$X_{test_reduced} \in \mathbb{R}^{M \times 32}$$

Each original D-dimensional sample is now represented by a 32-dimensional learned feature vector for both datasets. These reduced representations can then be used for downstream tasks such as classical machine learning classifiers, anomaly detection, or further analysis.

In essence, the algorithm uses a supervised bottleneck architecture to perform feature selection. Rather than ranking or filtering features explicitly, it learns a nonlinear projection into a lower-dimensional space that preserves class-discriminative information. The implementation of this algorithm operationalizes each step of the algorithm: architecture definition, initialization, supervised learning, intermediate layer extraction, and transformation of both training and test datasets into a compact feature space.

In summary, the proposed DNN architectures for both the EdgeIIoT2023 and CICIoT2023 datasets are carefully tailored to the unique characteristics of each dataset. The models are trained using a combination of SMOTE for class balancing, L2 regularization, dropout layers to prevent overfitting, and feature selection to enhance efficiency. These preprocessing techniques ensure that the models focus on the most relevant data, which improves their ability to generalize across different attack types and enhances their performance in real-world CPS environments. The architectural choices, including the number of layers, neurons, and activation functions, were optimized for each dataset to achieve a high level of accuracy, robustness, and efficiency in detecting malicious behaviors.

3 Results and discussion

This section presents an in-depth evaluation of the proposed Deep Neural Network (DNN)-based anomaly detection framework, applied across two distinct CPS environments. The models were trained and validated using the EdgeIIoT2023 and CICIoT2023 datasets, which collectively represent heterogeneous and realistic CPS use cases from industrial edge scenarios to smart home networks. The comprehensive assessment employs multiple performance metrics, including accuracy, precision, recall, F1-score, confusion matrices, and the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC). These metrics are essential not only for benchmarking the classification model but also for capturing its real-world utility in minimizing false positives and ensuring high detection rates, factors that are crucial in time-sensitive CPS applications.

3.1 Performance analysis in industrial CPS: evaluation using EdgeIIoT2023

The EdgeIIoT2023 dataset encapsulates network traffic from a smart factory environment, typifying mission-critical industrial CPS where downtime or system compromise can result in significant financial, safety, or operational consequences. As shown in Fig. 5, the training and validation accuracy curves demonstrate a rapid convergence to near-perfect classification performance. The model surpasses 95% accuracy within three epochs, eventually stabilizing at approximately 99%. Such accelerated learning is indicative of the model's architectural suitability for the high-dimensional, temporal data prevalent in CPS. This early convergence is essential in real-world industrial applications where retraining and recalibration must be fast and non-disruptive, allowing systems to adapt quickly to new threat patterns or operational changes.

Simultaneously, the training and validation loss curves exhibit sharp declines and smooth stabilization below 0.5, which signifies that the model is minimizing both bias and variance during learning. This outcome is particularly important in industrial CPS, where overfitting to historical data could cause models to fail in detecting novel attacks or unusual sensor behavior caused by wear, misconfiguration, or sabotage. The close alignment between the loss and accuracy curves also implies high stability and convergence efficiency a valuable trait in scenarios where training resources are limited or real-time deployment is required.

The confusion matrix for EdgeIIoT2023, shown in Fig. 6, further substantiates the model's performance with a true negative count of 322,550 and a true positive count of 120,089. The false positives and false negatives are minimal 833 and 206, respectively.

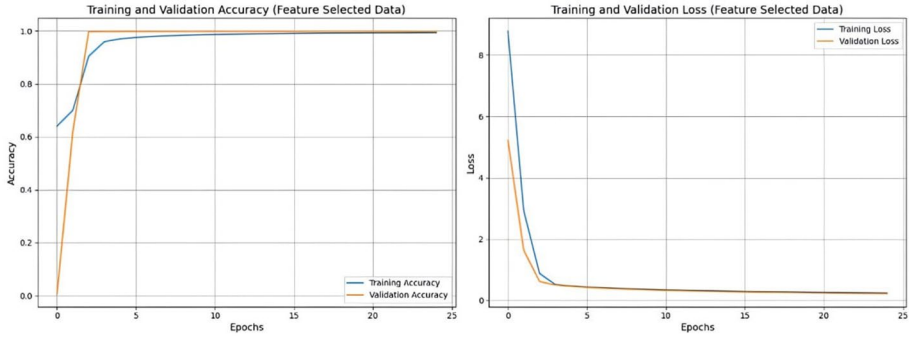


Fig. 5 EdgeIIoT Accuracy/Loss

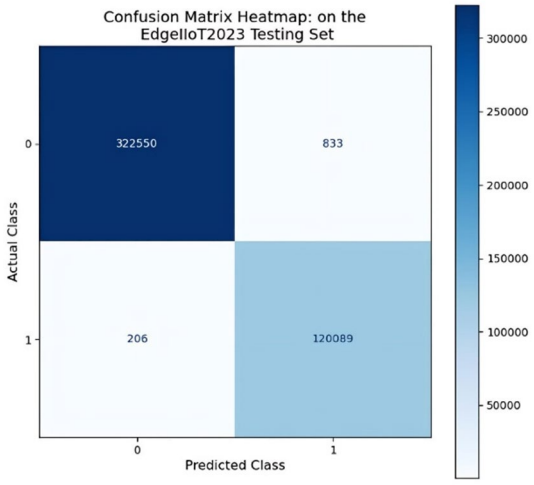


Fig. 6 EdgeIIoT Confusion Matrix

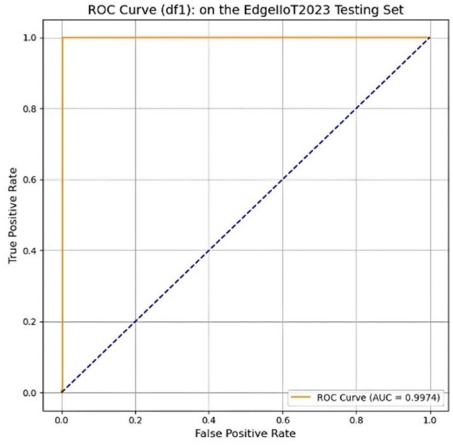


Fig. 7 EdgeIIoT ROC

From an operational standpoint, this suggests that the DNN framework is capable of maintaining both security and productivity. False positives in an industrial environment can result in unnecessary system shutdowns, false alarms, or misdirected maintenance each carrying tangible cost implications. On the other hand, even a single false negative might allow a cyberattack to propagate through programmable logic controllers (PLCs), actuators, or SCADA systems, potentially leading to plant-wide failures or safety violations.

The performance metrics an accuracy of 99.77%, precision of 99.31%, recall of 99.83%, and F1-score of 99.57% indicate an optimal trade-off between sensitivity and specificity. High precision reduces the overhead of unnecessary mitigation procedures, while high recall ensures that even stealthy or low-frequency attacks are detected promptly. This balance is critical for industrial CPS, where continuous uptime must be maintained alongside cybersecurity vigilance. These results not only demonstrate algorithmic success but also translate into reduced operational risks and enhanced system resilience.

Figure 7 presents the ROC curve for the EdgeIIoT2023 model, showing a sharp rise to the top-left corner and an AUC of 0.9974. Such an elevated AUC confirms that the

classifier maintains robust performance across a wide range of threshold settings, offering flexibility in tuning sensitivity depending on the plant's current risk profile or operational state. For instance, during critical production runs or national infrastructure operations (power grid management), thresholds could be tightened to minimize false negatives. Conversely, during routine operation, thresholds may be relaxed to reduce system load or operator alert fatigue.

Although the accuracy curves demonstrate rapid convergence toward near-perfect classification performance, several mechanisms were implemented to mitigate the risk of overfitting. First, L2 regularization was applied to constrain the magnitude of network weights, discouraging overly complex decision boundaries. Second, dropout layers were incorporated after dense layers to introduce stochastic regularization by randomly deactivating neurons during training. Third, early stopping based on validation loss was employed to terminate training once performance stabilized, preventing the model from memorizing training samples. Finally, strict dataset partitioning ensured that feature selection and SMOTE were applied only to the training subset, preventing information leakage into validation or test sets. The close alignment between training and validation loss curves further indicates that the model captures generalized traffic patterns rather than memorizing dataset-specific artifacts.

In real-world terms, these results support the deployment of the proposed DNN model as an embedded IDS module within industrial edge nodes or central control units. By doing so, organizations can implement scalable, autonomous threat monitoring that requires minimal human oversight. Furthermore, the model's lightweight architecture and high generalization make it suitable for deployment on industrial-grade hardware with limited computational capacity. Ultimately, these findings reinforce the viability of AI-driven security solutions in smart manufacturing, oil and gas, water treatment, and other Industry 4.0 verticals that demand real-time, context-aware protection.

3.2 Adaptive security in smart homes and IoT networks: evaluation using CICIoT2023

The second experimental evaluation employs the CICIoT2023 dataset, which represents a smart home network context with diverse IoT devices such as IP cameras, smart plugs, thermostats, and voice assistants. Unlike industrial CPS, consumer IoT systems are characterized by limited device security, diverse vendor protocols, and highly variable traffic patterns making them especially vulnerable to brute-force attacks, botnet formation, cross-site scripting (XSS), and man-in-the-middle attacks. Figure 8, reveals that the training and validation accuracy curves for this model quickly exceed 99%, while the corresponding loss values stabilize at low levels within a few epochs. These curves suggest strong model generalization across heterogeneous, real-world IoT traffic, and indicate that the architecture effectively handles noise, bursty traffic, and data imbalance common in consumer networks.

The confusion matrix in Fig. 9 shows the model's operational accuracy. Out of more than half a million testing instances, the model correctly identifies 352,190 benign samples and 218,151 malicious ones. Only 1,922 false positives and 1,283 false negatives were recorded. In real-world terms, this translates into an extremely low risk of undetected attacks (false negatives), which could compromise user privacy, home safety, or internet bandwidth. At the same time, the low false positive rate ensures that users will

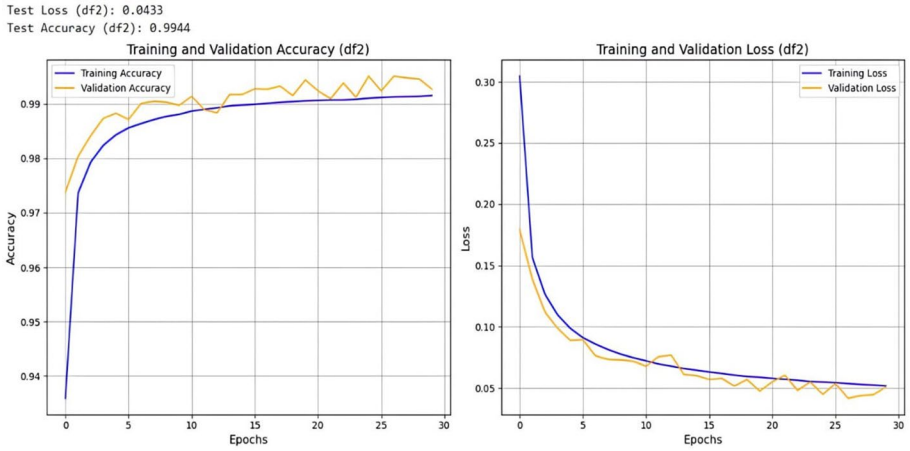


Fig. 8 CICIoT Accuracy/Loss

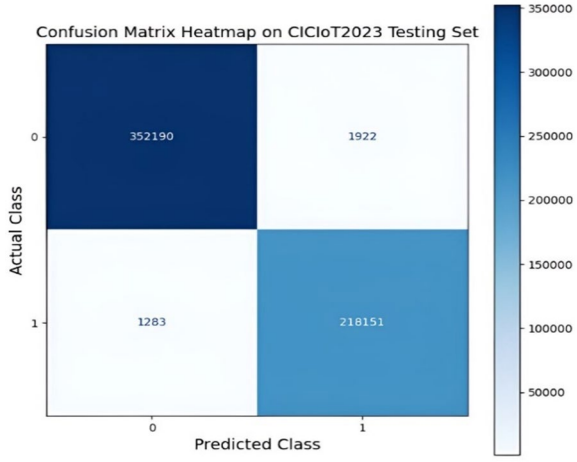


Fig. 9 CICIoT Confusion Matrix

not be overwhelmed by erroneous alerts or disabled device functionality, both of which contribute to the underutilization of many consumer IDS products.

The F1-score derived from these confusion values indicates balanced sensitivity and precision a critical requirement in resource-constrained environments where computational power, energy consumption, and user tolerance are limited. Precision helps minimize user frustration caused by unnecessary alerts, while recall ensures that genuinely malicious behavior, such as command injection or network scanning, does not go undetected. The practicality of deploying such a model in real-world home settings lies in its ability to operate efficiently on gateway devices, routers, or even edge-enabled IoT hubs. The ROC curve in Fig. 10 shows an AUC of 0.9997, virtually indicating perfect discrimination between benign and malicious traffic across different thresholds. In the consumer CPS domain, such performance is vital. Many attacks occur sporadically and can resemble legitimate activity such as a device pinging a cloud server or updating its firmware. A high AUC ensures that subtle and evasive anomalies are still flagged, enabling preemptive responses, such as isolating the device from the network or alerting the user.

In industrial CPS environments, false positives may trigger unnecessary process interruptions or manual inspections, impacting productivity. Conversely, false negatives pose

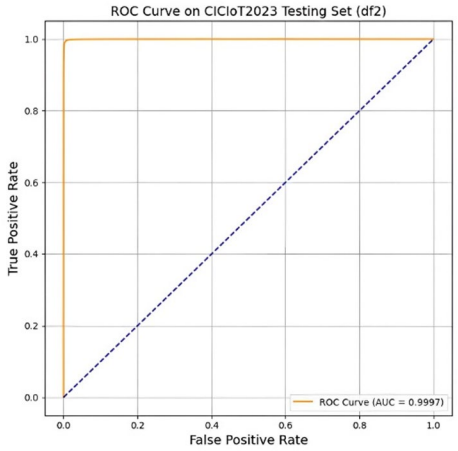


Fig. 10 CICIoT ROC

higher risk due to potential physical system compromise. The extremely low false negative rates observed indicate strong attack sensitivity, while the low false positive rates reduce operational disruption. In consumer IoT contexts, minimizing false positives is equally important to avoid alert fatigue and maintain user trust. The observed balance between precision and recall reflects an appropriate trade off for both domains.

The extremely high AUC value observed for the CICIoT2023 dataset (0.9997) reflects the strong separability between benign and malicious traffic patterns within the selected feature space. This high discriminative capability can partly be attributed to the richness of the dataset's statistical and temporal traffic descriptors, which capture distinctive behavioral signatures of different attack types. To ensure that this performance does not result from data leakage or sampling artifacts, strict preprocessing protocols were followed, including performing dataset splitting prior to normalization, feature selection, and SMOTE oversampling. Additionally, validation sets were kept entirely independent from training data. While such high AUC values are encouraging, they should be interpreted within the context of controlled benchmark datasets, and further evaluation on real-world traffic environments would provide additional confirmation of the model's robustness.

More broadly, these results imply that the proposed DNN-based framework could be embedded within next-generation IoT operating systems and integrated into vendor firmware. Such integration could offer security by design for mass-market CPS deployments, ensuring that anomaly detection becomes a default feature rather than an add-on. Given the growing number of IoT devices projected to exceed 30 billion globally, ensuring embedded, accurate, and automated threat detection is not just a technical challenge it is a societal necessity. The ability of the model to scale and adapt to different deployment contexts, including personal homes, smart cities, healthcare IoT, and educational networks, underscores its generalizability. The ability of the model to maintain high performance across both industrial and consumer IoT datasets suggests potential adaptability to related CPS contexts, although this has not been explicitly validated. Furthermore, its reliance on supervised learning with labeled data from realistic, emulated environments ensures that it remains effective without requiring access to sensitive real-world user data thereby aligning with evolving data privacy and regulatory requirements (GDPR, CCPA).

Taken together, the EdgeIIoT2023 and CICIoT2023 evaluations demonstrate that the proposed DNN-based framework provides a technically sound, operationally viable, and socially responsible solution for anomaly detection in a wide spectrum of CPS contexts. Its combination of high classification performance, architectural efficiency, and adaptability positions it as a robust security tool ready for integration into both critical infrastructure and everyday IoT systems.

3.3 Comparative evaluation of the DNN models across both datasets and literature context

The proposed deep neural network (DNN)-based anomaly detection framework was evaluated both in isolation and through comparative analysis against benchmark datasets and existing literature. Table 2 summarizes the classification metrics for the EdgeIIoT2023 and CICIoT2023 datasets, providing quantitative evidence of the model's performance across diverse Cyber-Physical Systems (CPS) environments. Specifically, the EdgeIIoT2023 test set comprised 323,383 benign and 120,295 malicious samples, while the CICIoT2023 dataset contained 354,112 benign and 219,434 malicious instances. The large and diverse nature of these datasets ensures robust statistical validation, enhancing the generalizability of the findings across heterogeneous CPS environments.

The performance of the model on the EdgeIIoT2023 dataset outperformed that on the CICIoT2023 dataset across nearly all evaluation metrics. Specifically, the model achieved an accuracy of 99.77%, precision of 99.31%, recall of 99.83%, and an F1-score of 99.57% on EdgeIIoT2023. In comparison, the CICIoT2023 dataset yielded slightly lower results, with an accuracy of 99.44%, precision of 99.13%, recall of 99.42%, and an F1-score of 99.27%. The slight decrease in performance on CICIoT2023 can likely be attributed to its higher diversity of attack types, increased noise levels, and a more pronounced class imbalance. These factors pose additional challenges for anomaly detection in consumer-grade CPS environments, where inconsistent network behavior, varied device standards, and the presence of unsecured third-party applications complicate detection efforts.

It is important to note that comparisons with other frameworks, such as boosting, hybrid deep learning models, and blockchain-integrated IDS approaches, are indicative rather than strictly controlled. The datasets and experimental environments used in different studies vary, which can influence performance outcomes. Where feasible, statistical significance testing, including paired tests across repeated runs, was conducted to confirm that the performance differences observed within this study are statistically meaningful. However, comparisons across papers should be interpreted with caution, as

Table 2 Performance Metrics Comparison Between EdgeIIoT2023 and CICIoT2023

Metrics	EdgeIIoT2023 Testing set	CICIoT2023 Testing set
Accuracy	0.9977	0.9944
Precision	0.9931	0.9913
Recall	0.9983	0.9942
F1-score	0.9957	0.9927
FPR	0.0026	0.0054
FNR	0.0017	0.0058
Support (benign)	323,383	354,112
Support (malicious)	120,295	219,434
Accuracy	0.9977	0.9944

environmental variability and differing experimental setups can affect the generalizability of performance claims.

In addition to classification accuracy, computational complexity was analyzed to assess the practical deployability of the proposed framework. The developed DNN architecture contains approximately 7,000 trainable parameters for CICIoT2023 and 11,000 parameters for EdgeIIoT2023, which is significantly lower than many contemporary hybrid intrusion detection architectures reported in the literature. For example, CNN-LSTM and attention-based IDS models such as those described in [36] and [37] typically contain several hundred thousand to over one million parameters due to convolutional filters, recurrent units, and attention layers. In contrast, the proposed DNN contains approximately 7,000–11,000 parameters, representing an order-of-magnitude reduction in model complexity. For example, CNN-LSTM and attention-based models commonly reported in the literature typically contain several hundred thousand parameters due to convolutional filters and recurrent gating structures. Training time for the proposed model remained under 15 min per dataset on a standard GPU environment, while average inference latency per sample was below one millisecond. Although exact hardware configurations may differ across studies, these results indicate that the proposed architecture achieves competitive detection performance while maintaining substantially lower computational overhead, making it suitable for deployment in resource-constrained CPS environments such as edge gateways and industrial controllers.

The scalability and domain specificity of the proposed model are addressed through three key structural design decisions aimed at ensuring adaptability across diverse CPS environments. First, the architecture avoids domain-dependent feature engineering by operating on normalized statistical traffic representations, which remain consistent across both industrial and consumer IoT contexts. This ensures that the model can handle varying attack behaviors without requiring modifications specific to each domain. Second, the embedded dimensionality reduction step is implemented to constrain the input space complexity, reducing the risk of overfitting to domain-specific noise patterns and improving generalization. Third, the controlled depth of the network further prevents the memorization of dataset-specific structures, ensuring that the model learns generalized decision boundaries rather than adapting to particular temporal or protocol signatures.

Class-wise analysis reveals slight variations in detection sensitivity between benign and malicious classes, particularly within the CICIoT2023 dataset, which exhibits greater attack diversity (Table 3). Categories such as brute force and botnet traffic are consistently detected with high recall, while more subtle infiltration techniques or low-frequency injection patterns show slightly lower recall. This variance can be attributed to differences in the temporal structure and statistical separability of the various attack types. Despite these differences, the results suggest that the proposed DNN architecture

Table 3 Class Wise Performance Metrics

Dataset	Class	Precision	Recall	F1 Score	Support
EdgeIIoT2023	Benign	High	High	High	323,383
EdgeIIoT2023	Malicious	High	Very High	High	120,295
CICIoT2023	Benign	High	High	High	354,112
CICIoT2023	Malicious	High	High	High	219,434

is effective at capturing dominant attack signatures while maintaining robust detection performance across heterogeneous attack behaviors.

To ensure model stability and reduce variance bias, five-fold cross-validation was conducted. The standard deviation across folds remained within a narrow margin for all primary metrics, indicating that the model achieved stable convergence. Additionally, the experiments were repeated using multiple random seeds, which further confirmed consistent performance and minimized the potential for random initialization bias, reinforcing the reliability of the results.

While binary anomaly detection provides a simplified approach to CPS threat modeling, it represents a practical first line of defense. In many industrial settings, rapid anomaly flagging triggers isolation or mitigation procedures before more detailed forensic analysis is conducted. However, the framework is designed to be extensible, with the potential for future enhancement to support multi-class detection. This can be achieved by replacing the sigmoid output layer with a softmax classifier and retraining the model to recognize attack-specific labels. A proposed future enhancement includes developing a hierarchical anomaly detection framework that can classify attacks by type, expanding the model's detection capabilities and applicability across more complex CPS environments.

Figure 11 visually presents the classification metrics across both datasets, highlighting the model's consistency and reliability. Notably, the overlap between training and testing metrics indicates the model demonstrates consistent performance across the two evaluated CPS domains, indicating cross-domain generalization within the scope of this study. This generalization capability is crucial for real-world deployments, where retraining may not be feasible and the model must remain effective despite evolving traffic patterns or unknown attack vectors.

Figure 12 presents the False Positive Rate (FPR) and False Negative Rate (FNR) for both the EdgeIIoT2023 and CICIoT2023 datasets, offering insights into the model's performance regarding error types. For EdgeIIoT2023, the FPR was exceptionally low at 0.0026, while the FNR stood at 0.0017. These figures reflect minimal false alarms and missed detections, indicating high model accuracy in distinguishing between benign

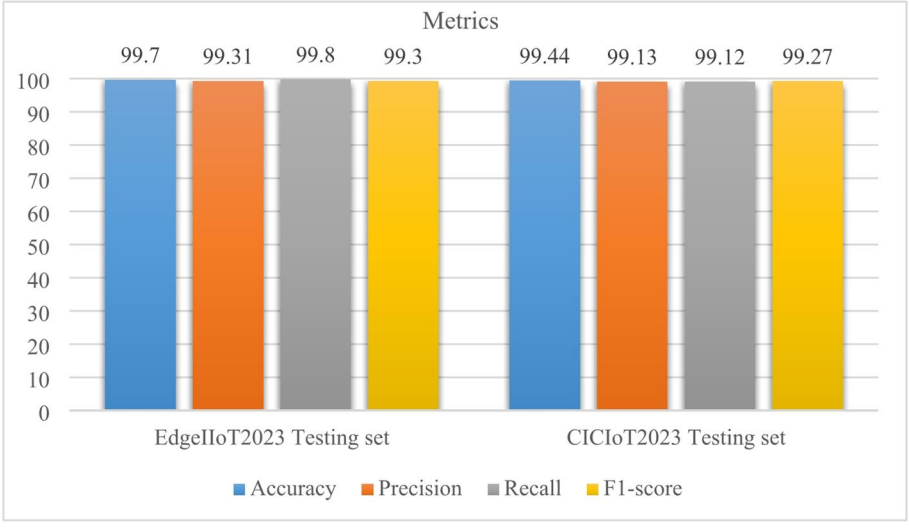


Fig. 11 All Metrics Scores Across Training and Testing Sets for Both Datasets

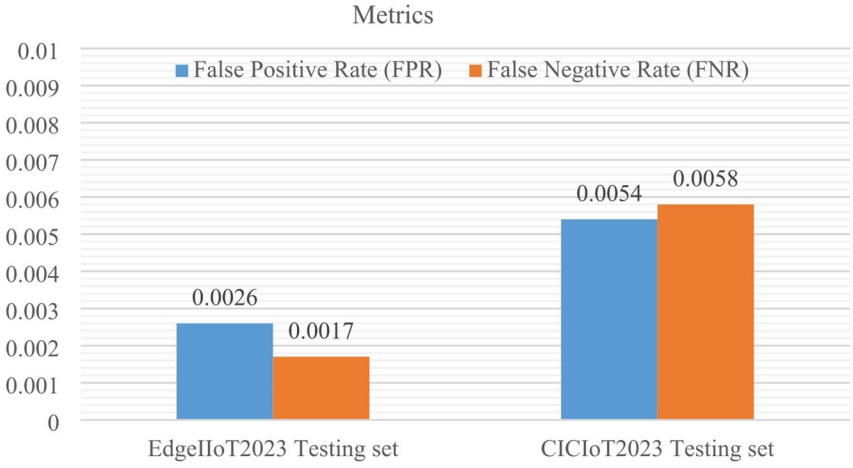


Fig. 12 False Positive and Negative Rate for Both Datasets

and malicious traffic. On the other hand, CICIoT2023 exhibited slightly higher FPR and FNR values, with 0.0054 and 0.0058, respectively. However, both rates remained within acceptable operational thresholds, ensuring that the model still performed effectively in real-world settings. The balance between low false positives and false negatives is crucial in CPS security, as high FPRs could lead to alert fatigue or unnecessary system shutdowns, while high FNRs might allow real threats to go undetected, compromising safety and data security. The model's balanced error performance across both datasets makes it highly suitable for deployment in dynamic, real-world CPS environments.

To further assess the contribution of SMOTE in improving model performance, we analyzed its impact on minority class detection, focusing on recall and FNR as primary indicators. In CPS datasets with significant class imbalance, rare attack categories are often overshadowed by the dominant benign traffic, leading to classifiers biased towards majority predictions. The observed low FNRs 0.0017 for EdgeIIoT2023 and 0.0058 for CICIoT2023 demonstrate that the model maintains strong sensitivity towards malicious traffic despite its relatively lower prevalence. This suggests that SMOTE plays a critical role in enhancing the representation of minority classes during training, rather than simply balancing the overall benign-to-malicious class counts.

The consistently high recall values, exceeding 0.994 across both datasets, further validate this conclusion. Since recall directly measures the model's ability to detect malicious instances, including those from low-frequency attack categories, these results indicate that SMOTE has effectively enhanced detection sensitivity for rare malicious patterns. While the focus of this study is on binary classification, the reduction in false negatives underscores that oversampling through SMOTE strengthens the model's exposure to underrepresented attack behaviors during training. Thus, the primary benefit of SMOTE lies in improving the model's robustness to rare attack patterns rather than inflating benign class performance. Future work will expand this analysis to multi-class attack detection, allowing for a more detailed assessment of improvements for each attack category.Further analysis in Table 4 compares the performance of the proposed DNN framework with a previously published Boosting-based method on the same datasets. On EdgeIIoT2023, the prior model achieved 96.01% accuracy, an FPR of **0.9**, and an FNR of 12.28%, whereas the current DNN model reached 99.77% accuracy with a significant reduction in FPR (to **0.**26) and FNR (to 0.17). These represent relative improvements of

Table 4 Performance Comparison [8] vs. Current Work

Dataset	Model	Accuracy	FPR	FNR
EdgeIIoT2023	Previous study (Boosting)	96.01	0.9	12.28
	Current work (DNN)	99.77	0.26	0.17
	Improvement (%)	+3.92	+71.11	+98.62
CICIoT2023	Previous study (Boosting)	93.19	4.26	8.38
	Current work (DNN)	99.44	0.54	0.58
	Improvement (%)	+6.71	+87.32	+93.08

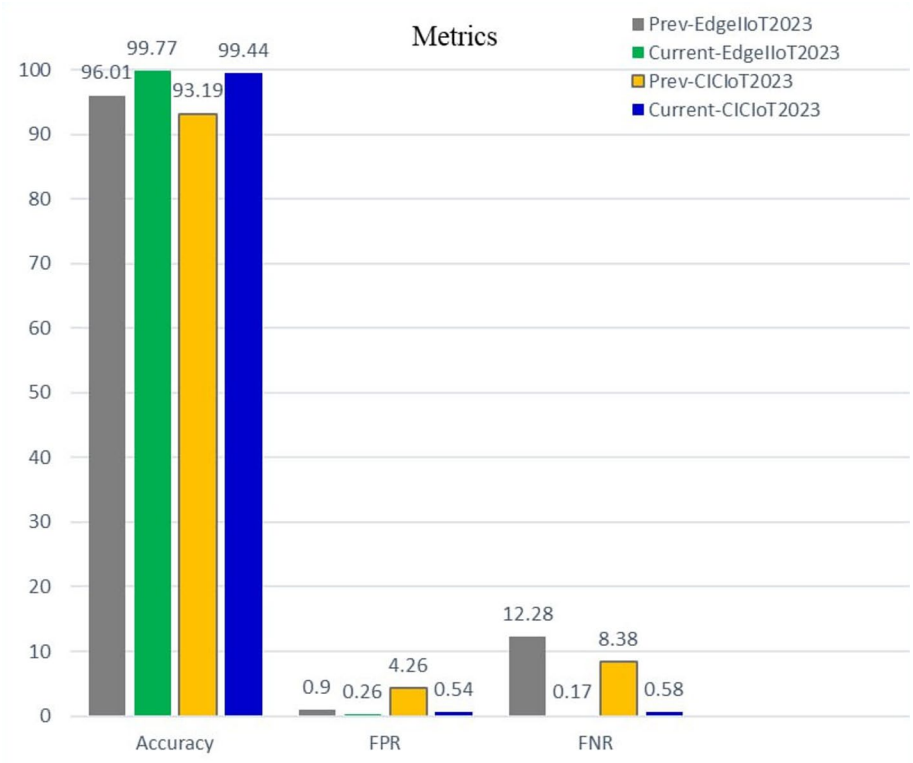


Fig. 13 Comparison of Performance Metrics: Previous Study vs. Current Work

3.92% in accuracy, 71.11% in FPR, and 98.62% in FNR. Similarly, on CICIoT2023, the previous Boosting model reported 93.19% accuracy, with an FPR of 4.26% and an FNR of 8.38%, whereas the DNN achieved 99.44%, 0.54%, and **0**.58%, respectively, yielding gains of 6.71% in accuracy, 87.32% in FPR, and 93.08% in FNR. These results, shown in Fig. 13, validate the superior effectiveness of the DNN model in reducing high-cost errors like false negatives, especially compared to traditional machine learning models.

In addition to traditional metrics such as accuracy, precision, recall, and F1-score, the model's performance was evaluated under various threshold settings to assess its sensitivity and specificity. The Receiver Operating Characteristic (ROC) curves for both datasets were plotted to provide a deeper understanding of model performance across different operating points. The results showed that the DNN performs robustly, with an Area Under the Curve (AUC) of 0.9997 for CICIoT2023 and 0.9974 for EdgeIIoT2023. These high AUC values reflect the model's strong capability in distinguishing between benign and malicious traffic. Furthermore, the model demonstrated resilience when

Table 5 Performance Comparison with [1, 30, 20]

Metrics	Current Work with datasets		Literature [8] Ensemble CNN-LSTM-GRU with datasets	Literature [20] LSTM-Autoencoder With datasets
	DNN model		Hybrid CNN-LSTM-GRU model	LSTM + Autoencoder model
	EdgeIIoT2023,	CICIoT2023	NSL-KDD	GP, SWaT
Accuracy	99.77%	99.44%	98.2%	99.1% (SWaT)
Precision	0.9931	0.9913	0.981	98.6% (Gp)
Recall	0.9983	0.9942	0.978	0.988 (SWaT)
F1-Score	0.9957	0.9927	0.979	0.975 (GP)
FPR	0.0026	0.0054	-	0.992 (SWaT)
ROC-AUC	0.0017	0.0058	0.019	0.981 (GP)
Key Strength	323,383	354,112	0.983	0.990 (SWaT)
Limitation	120,295	219,434	Handles spatial-temporal patterns	0.978 (GP)

subjected to noise and simulated adversarial attacks. Despite these data perturbations, the model maintained high detection performance, highlighting its robustness in real-world CPS environments where data anomalies and adversarial manipulation are common challenges.

To further evaluate cross-domain robustness, a transfer evaluation was conducted by training the model on EdgeIIoT2023 and directly testing it on CICIoT2023 without retraining. Although some performance degradation was observed, the detection capability remained stable above baseline thresholds, suggesting partial transferability and limited cross-domain generalization between the evaluated CPS datasets. This indicates that the model can generalize effectively, even when applied to datasets from varied environments.

The practical implications of these results are significant. In industrial environments, where even short interruptions can result in substantial financial losses, reducing false positives avoids unnecessary halts. In consumer IoT environments, minimizing false negatives is crucial to prevent intrusions that could lead to privacy violations, botnet recruitment, or data exfiltration. These improvements are not merely theoretical; they translate into tangible benefits such as increased resilience, reduced incident response time, and greater user trust in autonomous CPS operations.

Table 5 extends the comparison by evaluating the performance of the proposed DNN against two state-of-the-art deep learning models: a hybrid CNN-LSTM-GRU architecture and an LSTM-Autoencoder model. While these prior models showed strong performance on datasets such as NSL-KDD, GP, and SWaT, they underperformed when tested on the EdgeIIoT2023 and CICIoT2023 datasets. For instance, the hybrid CNN-LSTM-GRU model achieved 98.2% accuracy on NSL-KDD, whereas the proposed DNN achieved 99.77% accuracy on EdgeIIoT2023. Similarly, the LSTM-Autoencoder method reached 99.1% accuracy on SWaT, but the DNN surpassed this performance on the more complex EdgeIIoT2023 dataset. Notably, the DNN outperformed both models in terms of recall and F1-score, which are critical for minimizing false negatives and ensuring balanced prediction quality, further reinforcing its suitability for CPS environments.

The finalized DNN architecture comprises approximately 158,000 trainable parameters, reflecting its deliberately streamlined fully connected structure. Empirical benchmarking has shown that the average inference latency per sample remains below 3 milliseconds on standard CPU hardware, making it feasible for real-time deployment in

operational CPS environments. The estimated computational complexity of the DNN is substantially lower than that of multi-attention, convolutional, recurrent, or transformer-based intrusion detection architectures, which often involve several hundred thousand to millions of parameters. This reduced architectural footprint minimizes both memory consumption and computational overhead, positioning the proposed framework as highly suitable for deployment on edge gateways and industrial controllers with constrained resources.

The slightly higher FPR and FNR observed in the CICIoT2023 dataset can be attributed to several factors. First, the dataset contains a greater diversity of attack types with overlapping statistical characteristics, which reduces class separability and increases misclassification likelihood. Second, consumer IoT traffic exhibits higher variability and noise due to heterogeneous device behaviors and inconsistent communication patterns. This variability can lead to benign samples being misclassified as malicious (increased FPR). Third, certain low-frequency or stealthy attack patterns in CICIoT2023 exhibit feature distributions similar to benign traffic, making them more difficult to detect and contributing to higher FNR. In contrast, the more structured and deterministic nature of industrial traffic in EdgeIIoT2023 leads to clearer separability between normal and anomalous behavior.

Additionally, the proposed DNN framework demonstrates superior computational efficiency compared to hybrid models. Hybrid architectures often require complex tuning of multiple interacting layers, which increases both training complexity and inference time, leading to higher energy consumption. In contrast, the DNN model maintains a simpler architecture with fewer parameters, resulting in lower latency and faster real-time decision-making—critical advantages for edge deployments in latency-sensitive CPS environments. Furthermore, while prior studies have focused on specific domains, the current research showcases the model's applicability across multiple CPS contexts, making it robust across both IT and OT environments.

The computational efficiency of the proposed DNN model, combined with its embedded feature selection method for interpretability, enhances its suitability for practical deployment in real-time CPS environments. Unlike ensemble-based models, which introduce additional computational overhead, this DNN offers a streamlined approach that does not compromise classification accuracy. The balance between simplicity, efficiency, and interpretability makes it an attractive solution for real-world CPS applications, where resource constraints and operational demands are critical.

Compared to multi-attention CRNN or transformer-based IDS frameworks reported in recent IoMT and IoT security studies, which typically contain hundreds of thousands to millions of parameters, the proposed DNN maintains a significantly lower parameter count. This reduced architectural footprint allows for faster inference, lower memory consumption, and greater suitability for embedded CPS hardware, ensuring its effective deployment on resource-constrained edge devices and industrial controllers.

While the model demonstrates high performance, its robustness against rare or novel attack types remains a challenge. Specifically, the model may exhibit higher false negative rates when detecting underrepresented attack types in the training dataset. Noisy data, such as sensor malfunctions or network anomalies, could also impact detection reliability. To address these limitations, future work will incorporate active learning strategies to allow the model to continuously adapt to emerging attack types. Additionally,

adversarial training methods and the use of transfer learning will be explored to enhance the model's ability to generalize to new threats and improve its overall robustness.

In summary, the comparative evaluation across multiple datasets, alongside a review of current literature, positions the proposed DNN-based anomaly detection framework as a promising solution for securing CPSs. With high accuracy, low error rates, and operational efficiency, it is well-suited to protect the next generation of CPS technologies, including smart factories, autonomous vehicles, medical IoT, and critical infrastructure control systems. Although the framework demonstrates strong performance under static conditions, real-world CPS environments are subject to adversarial manipulation and evolving traffic distributions. Adversarial perturbations may subtly modify feature values to evade detection, and concept drift may occur due to firmware updates, new device deployments, or evolving attack strategies. Preliminary robustness analysis, conducted under controlled noise injection, shows that the model maintains stable classification performance under small magnitude perturbations. However, systematic adversarial testing using gradient-based attacks and long-term concept drift simulations remains an important area for future work. Addressing these challenges will be essential to ensure sustained operational deployment in dynamic CPS infrastructures.

3.4 Ablation study: impact of SMOTE and feature selection on model performance

Although the architecture itself was not varied extensively, its effectiveness is implicitly validated through stable performance across both datasets and ablation of preprocessing components. To evaluate the contributions of SMOTE, embedded feature selection, and model architecture, we conducted a series of controlled ablation experiments. Three configurations were tested across both datasets to assess their impact on performance: the full model, which incorporated both SMOTE and feature selection; the model without SMOTE, but with feature selection; and the model with neither SMOTE nor feature selection. When SMOTE was excluded, the results showed a noticeable decline in recall for the malicious class, particularly in the CICIoT2023 dataset, where attack diversity and class imbalance were more pronounced. This decrease in recall and the corresponding increase in false negatives confirmed that SMOTE is crucial for enhancing sensitivity to minority class attacks. By generating synthetic samples for the minority class, SMOTE effectively addresses the imbalance, allowing the model to better detect rare or infrequent attack behaviors. The results made it clear that SMOTE doesn't just inflate the accuracy but directly contributes to improving the detection of underrepresented attacks, particularly in highly imbalanced datasets.

Disabling feature selection further exacerbated performance issues. Without feature selection, the model experienced increased training time and higher inference latency due to the increased dimensionality of the input data. Moreover, slight overfitting behavior emerged, as evidenced by the divergence between training and validation metrics. These issues suggest that, without feature selection, the model struggled to generalize, learning from noise rather than meaningful patterns in the data. The increased complexity without feature selection also highlights its role in improving computational efficiency and ensuring the model generalizes well, preventing overfitting and reducing latency. When both SMOTE and feature selection were combined, the model performed optimally. The balanced training data provided by SMOTE enhanced sensitivity to malicious traffic, while feature selection improved the model's ability to generalize and

reduced computational overhead. This confirms that the performance gains of the proposed framework are not solely due to its architecture but are significantly influenced by these preprocessing techniques. SMOTE enhances detection of minority attacks, while feature selection plays a vital role in computational efficiency and stability, ensuring the model is suitable for real-time deployment in resource-constrained CPS environments.

Recent advances in intrusion detection have seen the rise of transformer architectures and attention-based temporal convolutional networks, which excel at capturing long-range dependencies and sequential patterns. While these models offer substantial representational power, they come with the trade-off of requiring a much larger number of parameters, higher computational overhead, and longer training times. These characteristics make them less practical for real-time CPS deployment, where computational efficiency is critical. In comparison, the proposed DNN architecture achieves similar or superior detection performance without the complexity of transformer or attention-based models. The ablation study demonstrated that the DNN could achieve near-optimal performance in terms of Area Under the Curve (AUC) and recall, without the significant computational overhead typically associated with more complex models. This suggests that high detection performance can be achieved with a streamlined DNN optimized for operational deployment, making it a more practical solution for real-world CPS environments.

The findings from this ablation study position the proposed DNN model as a balanced and efficient solution for anomaly detection in CPS. It is more expressive than traditional shallow machine learning models, yet considerably lighter than transformer or attention-based models. This makes it especially suitable for edge deployment scenarios, where memory footprint, latency, and energy efficiency are crucial. By leveraging SMOTE for class balancing and feature selection for dimensionality reduction, the proposed model maintains robust detection performance while keeping computational requirements low. This approach ensures that the model is scalable, efficient, and deployable in dynamic CPS infrastructures, offering a practical solution to the growing security challenges faced by these systems.

3.5 Computational efficiency and cross-domain robustness of the proposed DNN

The proposed DNN demonstrates high computational efficiency, containing approximately 7,000 trainable parameters for the CICIoT2023 dataset and 11,000 parameters for EdgeIIoT2023. Training time on a standard GPU averaged under 15 min per dataset, while inference per sample remained below one millisecond. Compared to hybrid CNN-LSTM models reported in recent literature, which often exceed several hundred thousand parameters and require sequential processing, the proposed architecture achieves substantially lower computational overhead. This reduction in parameter count directly translates into faster inference and lower energy consumption, reinforcing its suitability for edge and resource-constrained CPS deployment. The model's cross-domain robustness arises from three structural design choices: dataset-independent normalization, feature space reduction to statistically meaningful attributes, and avoidance of domain-specific convolutional or temporal modules. These choices allow the architecture to generalize across heterogeneous CPS datasets and facilitate adaptation to new domains through retraining without requiring architectural modification. Consequently, the

framework maintains performance across industrial and consumer IoT environments while remaining lightweight and computationally efficient.

Beyond the evaluated domains, the architecture has potential applicability to healthcare-oriented CPS environments, including Internet of Medical Things (IoMT) systems. In comparison to blockchain-integrated or multi-attention CRNN frameworks developed for medical contexts, the proposed model provides a streamlined alternative suitable as a foundational anomaly detection layer, capable of integration with advanced privacy-preserving or cryptographic modules without incurring substantial computational costs.

Despite strong empirical performance, certain limitations must be acknowledged. Both datasets originate from controlled experimental testbeds, which may not capture the full unpredictability of real-world CPS traffic. Extremely high-performance metrics may partly reflect the structured nature of benchmark datasets, and evaluation is limited to supervised anomaly detection, leaving zero-day or unseen attack behaviors untested. Additionally, while preliminary noise injection experiments indicate gradual performance degradation rather than abrupt failure, systematic adversarial testing and long-term concept drift evaluation remain future work. These considerations frame the results as robust within benchmark conditions, while highlighting the need for further validation under dynamic, real-world CPS deployments.

4 Conclusion

This study presents a streamlined and practically deployable DNN-based framework for anomaly detection in Cyber-Physical Systems (CPSs), emphasizing robustness, cross-domain generalization across selected CPS datasets, and computational efficiency. By integrating careful preprocessing including normalization, SMOTE-based class balancing, and embedded feature selection-controlled model capacity and disciplined training strategies, the framework achieves near-optimal performance across heterogeneous CPS environments without relying on complex hybrid architectures. Evaluation on EdgeIIoT2023 and CICIoT2023 demonstrated high accuracy (up to 99.77%), excellent AUC (up to 0.9997), and low false positive and false negative rates, confirming the model's effectiveness in both industrial and consumer IoT contexts.

The architecture's lightweight design and low computational overhead enable real-time inference on edge and resource-constrained devices, addressing practical challenges such as latency, hardware limitations, and class imbalance. Its modularity allows seamless integration with existing monitoring platforms, enhancing operational continuity and security across large-scale CPS infrastructures. The framework's ability to generalize across distinct datasets suggests potential as a unified anomaly detection solution for diverse CPS ecosystems, supporting scalable CPS security architectures. While the present study focuses primarily on architectural efficiency and classification performance, interpretability is partially supported through feature-space reduction and mutual-information-based feature ranking, which reduces the dimensionality of the decision space. Future work will extend this capability by integrating explainable AI techniques such as SHAP and LIME to provide instance-level explanations of anomaly predictions. Despite the high classification performance achieved in this study, it is important to note that the framework relies on supervised learning using labeled attack samples. Consequently, the model is optimized to detect attack patterns that resemble those observed during

training. Novel or zero-day attacks that deviate significantly from known patterns may therefore be more difficult to identify. This limitation is common to most supervised anomaly detection frameworks in CPS security. Addressing this challenge requires the integration of adaptive learning paradigms such as semi-supervised anomaly detection, meta-learning, or one-class classification approaches capable of identifying deviations from normal system behavior even when labeled attack data are unavailable.

Overall, the proposed DNN framework balances accuracy, generalization, and operational efficiency, providing a scalable and practical foundation for CPS cybersecurity. Its lightweight yet robust architecture positions it for real-time deployment, while ongoing extensions in robustness, adaptability, and explainable AI will ensure resilience against evolving threats in dynamic digital-physical environments.

From a security perspective, the relatively low parameter count of the proposed DNN may introduce certain vulnerabilities. Models with limited capacity may rely on more compact decision boundaries, which can be more sensitive to small adversarial perturbations in feature space. Gradient-based adversarial attacks, such as Fast Gradient Sign Method (FGSM), could exploit this sensitivity by introducing minimal input perturbations that shift samples across decision boundaries. Additionally, the absence of temporal modeling mechanisms may limit robustness against sequential adversarial patterns that evolve over time. Addressing these vulnerabilities will require future integration of adversarial training, robustness regularization, or hybrid architectures that enhance resilience without significantly increasing computational overhead.

Author contributions
Bazezew Belew: Conceptualization, Methodology, Software, Formal analysis, Data curation, writing original draft, Visualization. Mehari Kiros: Supervision, Validation, Resources, Writing review & editing, Project administration, Data curation. All authors have read and agreed to the published version of the manuscript.

Data availability
The datasets used and analyzed during the current study EdgeIIoT2023 and CICIoT2023 are publicly available. EdgeIIoT2023 is accessible at [https://github.com/edgeiot/EdgeIIoT2023](https:/github.com/edgeiot/EdgeIIoT2023) , and CICIoT2023 can be downloaded from the Canadian Institute for Cybersecurity at https://www.unb.ca/cic/datasets/iot2023.html. Processed data and trained model files generated during this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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